Why these particular markets? They are all highly liquid and easily traded, and that's the most important part. The typical trend-following futures fund would use around 100 different contracts and some make artificial markets by trading spreads between, for instance, gold and silver.

I use daily data only, not just to simplify but also because you can successfully run these types of strategies without bothering with intraday data. Since this is futures data, you need to make sure that the data is properly adjusted, as I discuss in Chapter 2. Don't skip or underestimate the importance of this detail. If your futures data is not properly adjusted, you are simply wasting your time building strategies and will be wasting your money trading them later on.

Position Sizing

Without a decent position-sizing formula it really does not matter how good the trading rules are. The instruments included in our portfolio above show very different volatility profiles, not to say extremely different volatility profiles. The term volatility is admittedly used loosely here and what we are concerned with is the potential price fluctuations of different instruments based on their recent past.

Although the Euro Stoxx 50 index can easily move 1–2% in a day and sometimes even 4–5% in a very eventful day, a move of just 0.5% in a day would be practically unheard of for Eurodollar. So if we were simply to put the same notional dollar amount in each trade, the portfolio would immediately be dominated by the volatile instruments and not much impact at all would come from the less volatile. This is not what we want to do, so instead we need to find ways to take the volatility of each instrument into account when deciding how many contracts to buy or sell in each market. There are several variations on how to do this: some prefer to base it on average true range, others on standard deviation and some may prefer to make their own formulas. In essence, the methodologies used by most CTAs are more or less the same, the principle being that you take larger sizes of less volatile instruments with the aim of having each position theoretically capable of making the same bottom line profit or loss impact on the overall strategy.

For our strategies in this book I use a method based on so-called average true range, which has been widely known for a long time and used for at least four decades. It aims to measure how big a normal daily move is for each instrument and then use that as a basis for position sizes. The true range refers